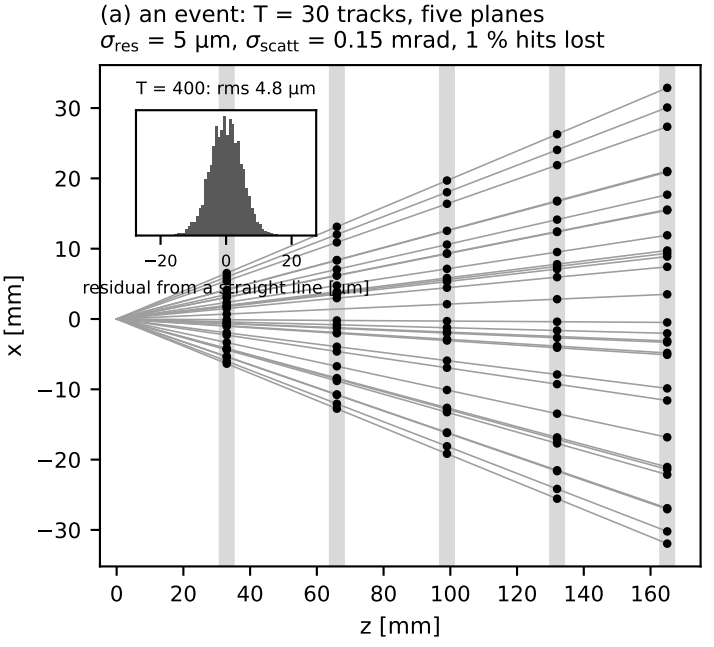
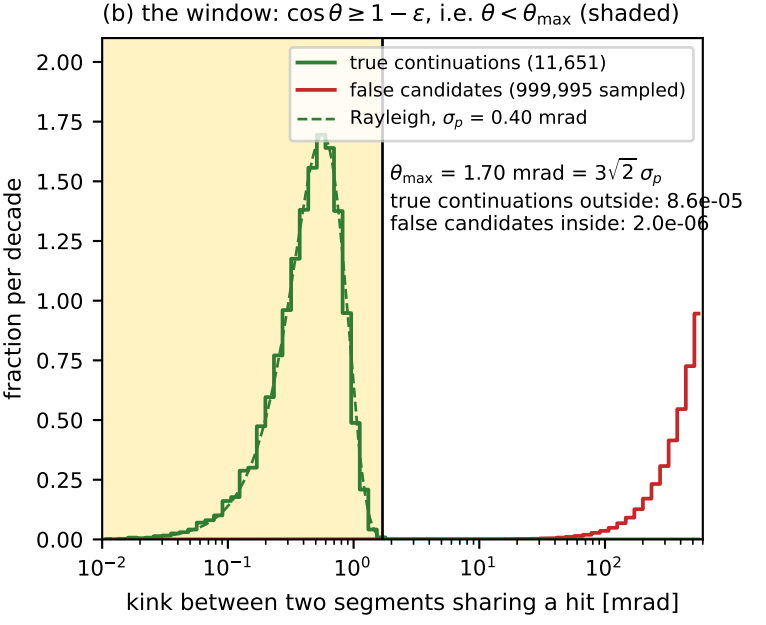
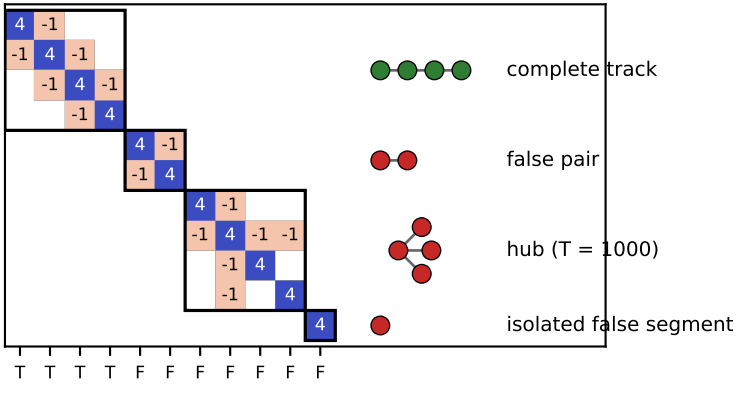
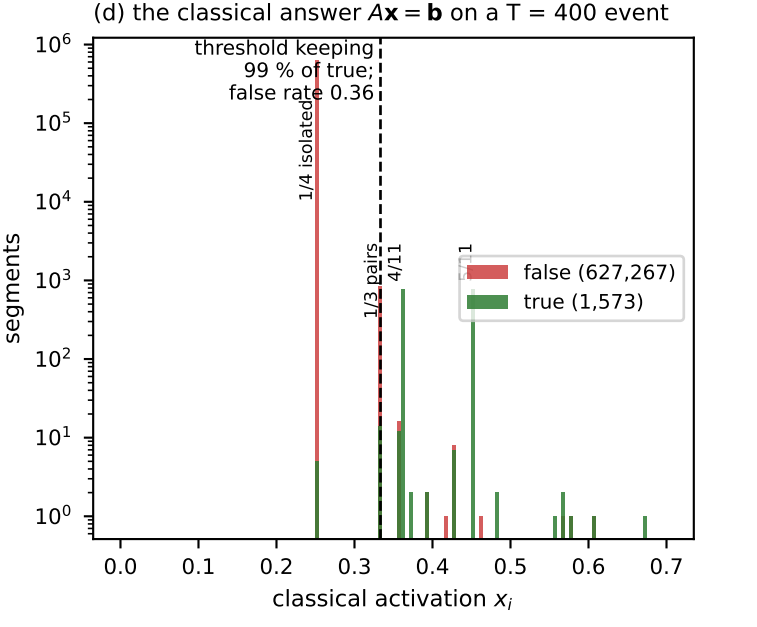

(a) an event: T = 30 tracks, five planes
σ_{res} = 5 µm, σ_{scatt} = 0.15 mrad, 1 % hits lost
T = 400: rms 4.8 µm
residual from a straight line [µm]
x [mm]
z [mm]


(b) the window: $\cos\theta \geq 1-\varepsilon$, i.e. $\theta < \theta_{max}$ (shaded)
true continuations (11,651)
false candidates (999,995 sampled)
Rayleigh, σ_p = 0.40 mrad
θ_{max} = 1.70 mrad = $3\sqrt{2}\,\sigma_p$
true continuations outside: 8.6e-05
false candidates inside: 2.0e-06
fraction per decade
kink between two segments sharing a hit [mrad]


(c) $A = (\alpha+\beta)I - C$ is block-diagonal: one block per group
of compatible segments, with its graph (green true, red false)
complete track
false pair
hub (T = 1000)
isolated false segment


(d) the classical answer $A\mathbf{x} = \mathbf{b}$ on a T = 400 event
threshold keeping
99 % of true;
false rate 0.36
1/4 isolated
1/3 pairs
4/11
1
false (627,267)
true (1,573)
segments
classical activation x_i